

Analyzing factors contributing to real-time train arrival delays using seemingly unrelated regression models

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ABSTRACT

Understanding the impact of various factors on train arrival delays is a prerequisite for effective railway traffic operating control and management. Existing studies analyze the train delay factors using a single, generic regression equation, restricting their capability in accounting for heterogeneous impacts of spatiotemporal factors on arrival delays as the train travels along its route. The paper proposes a set of equations conditional on the train location for analyzing train arrival delay factors at stations. We develop a seemingly unrelated regression equation (SURE) model to estimate the coefficients simultaneously while considering potential correlations between regression residuals caused by shared unobserved variables among equations. The railway data from 2017 to 2020 in Sweden are used to validate the proposed model and explore the effects of various factors on train arrival delays. The results confirm the necessity of developing a set of station-specific train arrival delay models to understand the heterogeneous impact of explanatory variables. The results show that the significant factors impacting train arrival delays are primarily train operations, including dwell times, running times, and operation delays from previous trains and upstream stations. The factors of the calendar, weather, and maintenance are also significant in impacting delays. Importantly, different train operating management strategies should be targeted at different stations since the impacts of these factors could vary depending on where the station is.

1. Introduction

The railway system is an important transport mode in many countries, but it faces great challenges in service punctuality given the aging infrastructure and increasing demand. The Swedish Transport Administration (Trafikverket) has a goal of 95% of all scheduled trains arriving at their respective destinations with less than five minutes of delays (Brisère, 2021). The punctuality of trains in Sweden reached 94% for the year 2020 during the pandemic (Trafikanalys, 2021), but dropped back to 90% in 2021 (close to the period 2013–2017 (Trafikanalys, 2022)). There is an urgent need to improve railway punctuality with the recovering demand after the pandemic.

Understanding the factors impacting train arrival delays is a prerequisite for effective operations controls and management in railways. Kecman and Goverde (2015) broadly categorized data-driven train delay models into global and local models. The global models utilize historical train operation data of the entire train network to investigate factors affecting train delays. The models are commonly used for strategic and tactical levels of railway traffic planning (Gorman, 2009; Marković et al., 2015). The local models utilize real-time data along particular train lines to continuously predict and update predictions in response to railway

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traffic evolution. The models are commonly used for proactive real-time (or operational) railway traffic management and passenger information (Li et al., 2021; Taleongpong et al., 2022; Wang and Zhang, 2019; Wen et al., 2020; Oneto et al., 2017, 2018; Lulli et al., 2018; Huang et al., 2020a,b). It is recognized that local models provide more accurate train arrival delay predictions.

This paper aims to understand the factors that influence real-time train arrival delays (local model). Existing studies analyze the contributing factors by modeling the train arrival delays using a single, general equation regardless of the train location, which is limited in capturing the heterogeneous impact of factors in the real-time train arrival delay analysis. Conceptually, the factors used for train arrival delay analysis will have different levels of influence in time and space as trains move towards the destination. In addition, most studies explore train operational factors, and very few studies were found on understanding the impact of a comprehensive set of factors due to data availability (see Table 2). Unlike the previous study, which focused solely on historical variables, this study examines the impacts of both real-time and historical factors on train arrival delays.

To address these gaps, the study proposes conditional regression equation models to understand the heterogeneous impact of a comprehensive set of factors (i.e., train operations, network, weather, maintenance, and calendar variables) on real-time train arrival delays. Case studies are conducted using selected railway lines in Sweden. The main contributions are:

- Propose a set of equations for analyzing real-time train arrival delays, conditional on the train's location. Different from existing studies, it captures the heterogeneous impact of station delay factors.
- Introduce the adoption of the real-time prediction concept when investigating the impacts of both historical and real-time explanatory variables on train arrival delays. Thus, this study ensures the adoption of the most up-to-date information for real-time train arrival delay factor analysis by leveraging the latest availability of data.
- Develop a seemingly unrelated regression equation (SURE) method to simultaneously estimate the coefficients of all equations by capturing the correlations of the equation residuals caused by the shared unobserved features in these equations.
- Identify and quantify the various important factors impacting real-time train arrival delays using data for Sweden's high-speed rail lines, and discuss potential implications for practice.

The rest of this paper is structured as follows: Section 2 presents a literature review of prior research. Section 3 gives a brief description of the real-time train arrival delay factors analysis problem and an introduction to the SURE model used for estimating the model parameters. Section 4 describes the case study of high-speed trains in Sweden, including the data preparation and results of the case study. Section 5 concludes the paper with the main findings and future work.

2. Literature review

Various models have been proposed to model and predict train delays. We review the data-driven train delay prediction models in two aspects: (1) Modeling approaches and (2) Explanatory variables.

2.1. Modeling approaches

According to Spanninger et al. (2022), data-driven train delay predictions are based on a direct prediction of delays at the desired station or point in time rather than explicit modeling of train-event dependency structure to capture traffic flow dynamics. Kecman and Goverde (2015) broadly categorized data-driven train delay models into global and local models.

The global model is often used to examine the effects of key factors on train delays, especially at the strategic and tactical levels of railway traffic planning. The major advantage of the global regression model is the generalizability of its results, that is, the results derived from the global model can be applied to train lines that are not in the training datasets (Kecman and Goverde, 2015; Chiou et al., 2015). Statistical regression models are commonly adopted to develop global train delay models. Gorman (2009) identified that primary congestion-related factors such as train meetings, passings, and overtakings have the largest impact on congestion delay. Few efforts have been devoted to developing accurate global train delay prediction models by using machine learning and deep learning models (Marković et al., 2015; Wang and Zhang, 2019). However, the major drawback of the global model is that time and geographical variation in the relations between variables are masked. This is because the global model ignores the existence of local variations due to the spatiotemporal autocorrelation by utilizing aggregated data and treating all observations independently.

In contrast to the global model, the local model takes into consideration spatiotemporal characteristics by modeling train delay variation along the train line over multiple stations with continuous inputs of real-time train operation data. The study from Bao et al. (2021) demonstrated that the importance of features used for train delay predictions changes in time and in space as the trains move towards the destination reflecting the importance of taking into consideration of spatiotemporal characteristics. More advanced methods that map the complex relationship between inputs and outputs are commonly adopted with high accuracy, such as conventional machine learning (Li et al., 2021; Taleongpong et al., 2022; Wang and Zhang, 2019), neural network (Wen et al., 2020; Oneto et al., 2017, 2018) and hybrid approach (Lulli et al., 2018; Huang et al., 2020a,b).

All these studies consistently show that the local model is developed to generate accurate train delay predictions for real-time traffic management instead of examining the key factors affecting train delays throughout the line. In addition, existing factor analysis models are designed for planning purpose and the single, general equation model is not applicable to the factor analysis in real-time train arrival delays. This paper proposes a set of equations for analyzing real-time train arrival delays, conditional on the train's location. It can also capture the heterogeneous impact of station delay factors, based on which informed countermeasures could be derived for effective traffic management.

Table 1
Categorization of explanatory variables in existing literature.

Variable	Type of feature	Example
Train operation	Train	Train type, train length, train tonnage, train speed, train horsepower, train count, train direction, train priority and train order
	Train events	Arrival delays, departure delays, scheduled dwell time, actual dwell time and dwell time delays, scheduled run time, actual run time and run-time delays, travel time at individual sections between stations, buffer time, actual headways, scheduled headways, headway delays, train interaction(meet, pass, overtaking)
Rail network	Location	Station attributes, areas attributes, zones, railway checkpoints, railway section (the number of stops/distance to the destination station), distance traveled, percent of the journey completed, and the number of stops/distance between consecutive stations
	Infrastructure	The number of tracks, track segment occupancy, track occupation conflict indicators, track allocation, platform conflict indicators, designated platform, platform change status, the availability of sidings and the number of sidings
External	Calendar	Times of day, days of the week, months of the year, holidays or working days, season, peak hours or off-peak hours
	Weather	Temperature, wind speeds, snow depths, rainfall or precipitation
	Disturbance	Period of delay occurrence, causes of disturbance, severity of disturbances, disturbances length, the number of affected trains
	Others	Planned maintenance, passengers, train crews

2.2. Explanatory variables

The explanatory variables are grouped into three main categories: (1) Train operation variables; (2) Network variables; and (3) External variables, as shown in Table 1.

In general, the train operation variables include train and train event variables. Train variables refer to variables that capture train traffic heterogeneity. Different train types have different train specifications, resulting in varying infrastructure capacity requirements, operation frequencies, and priorities. Longer trains, for instance, dwell longer due to the spatial distribution of alighting and boarding passengers, necessitating longer times for train operators to ensure no passenger boarding prior to departure (Li et al., 2016). Train event variables refer to variables that keep track of both scheduled train activities and actual train movements in real practice. Many studies found that train event variables are the primary factors affecting train delays (Huang et al., 2020a; Lee et al., 2016; Taleongpong et al., 2022; Bao et al., 2021; Shi et al., 2021). Several studies revealed that current arrival delays are the most significant factors when predicting one-station ahead train delays (Bao et al., 2021; Shi et al., 2021), but (Li et al., 2021) indicated that this is only true if the current delays are longer than three minutes. Many existing studies have demonstrated good performance of the train delay prediction models when the upstream stations' delays and upstream trains' delays are considered (Li et al., 2022; Huang et al., 2020a,b; Oneto et al., 2018, 2017).

Network variables include location, infrastructure, and disturbance variables. Location variables refer to variables that describe the position of the train in order to capture the pattern of train delays at the respective location. This is due to the fact that different locations along the train network have different train operation conditions, such as variations in infrastructure availability, topography, and demand for rail service. Marković et al. (2015) showed that train delays increased with the distance traveled but decreased with better infrastructure conditions along the route. Infrastructure variables encompass variables related to the structures, buildings, and equipment designed to support railway systems. Since infrastructures are shared facilities among trains, the inclusion of infrastructure variables allows train delays caused by infrastructure constraints or infrastructure faults to be tracked. According to Veiseth et al. (2007), infrastructure faults account for 30% of delay hours in Norway.

The common external variables include calendar, weather, and disturbance variables. Calendar variables intend to capture train delay distribution over different periods of time. For instance, trains dwell longer at stations to serve more passengers during peak hours (Li et al., 2016); arrival delays vary with seasons (Laifa et al., 2021), months of the year (Grandhi et al., 2021), and days of the week (Ceder and Hassold, 2015). Weather variables refer to variables that quantify the state of the atmosphere, such as temperature, wind speed, snow depth, and rainfall. Several studies showed that the inclusion of weather variables improved the accuracy of the train delay prediction models (Huang et al., 2020a; Oneto et al., 2017). Also, many studies show that extreme weather has significant negative impacts on train punctuality since severe weather conditions affect ordinary train operations (Brazil et al., 2017; Wang and Zhang, 2019; Zakeri and Olsson, 2017). The disturbance variables are variables that carry information about the disturbance and its implications. These variables provide insight into the consequences of delays on operations and offer optimum advisory strategies on real-time train dispatching, especially when the trains encounter disturbance. Li et al. (2021) found that delays less than three

Table 2

Categorization of the existing literature based on explanatory variable.

Author	Train operation variables										Network variables					External variables			
	Train variables			Train event variables							Location variables			Infra variables					
	Train characteristics	Train type	Train traffic	Arrival delays	Departure delays	Dwell times	Run/travel time	Headways	Buffer times	Train interaction	Distance features	Stations features	Area features	Track features	Platform features	Calendar	Weather	Disturbance	Others
Huang et al. (2019)				✓				✓										✓	
Wen et al. (2020)		✓		✓	✓		✓									✓			
Lulli et al. (2018)		✓	✓	✓		✓	✓				✓	✓				✓	✓		
Marković et al. (2015)		✓		✓			✓	✓			✓			✓	✓	✓			
Oneto et al. (2017)						✓	✓									✓	✓		
Li et al. (2021)				✓			✓				✓							✓	
Li et al. (2020b)			✓	✓			✓				✓	✓							
Huang et al. (2020a)				✓	✓	✓	✓	✓			✓	✓		✓			✓		
Taleongpong et al. (2020)			✓	✓	✓	✓	✓				✓	✓				✓			
Pongnumkul et al. (2014)				✓															
Yaghini et al. (2013)												✓	✓			✓			
Nair et al. (2019)	✓	✓		✓		✓	✓	✓			✓	✓		✓	✓	✓	✓		✓
Nabian et al. (2019)	✓		✓	✓			✓				✓					✓	✓		✓
Oneto et al. (2018)				✓		✓	✓									✓			
Wang and Zhang (2019)			✓														✓	✓	
Huang et al. (2020b)				✓	✓	✓	✓	✓			✓	✓		✓					
Gaurav and Srivastava (2018)	✓	✓	✓	✓							✓	✓	✓			✓			
Gorman (2009)	✓	✓	✓				✓	✓		✓									
Oneto et al. (2016)						✓	✓										✓		
Shi et al. (2021)	✓		✓	✓		✓	✓		✓		✓	✓		✓	✓				
Grandhi et al. (2021)								✓					✓	✓		✓	✓	✓	
Zhang et al. (2021)				✓												✓	✓		
Rößler et al. (2021)				✓	✓											✓	✓		
Huang et al. (2021)				✓	✓	✓	✓	✓			✓	✓		✓			✓		
Laifa et al. (2021)			✓	✓	✓						✓		✓			✓			
Bao et al. (2021)			✓						✓		✓								

minutes are most vulnerable to the period of delay occurrence. Li et al. (2020a) discovered that the average number of affected trains was related to the severity of the delays, which increased with higher average numbers.

Table 2 shows the categorization of existing literature. Based on Table 2, most studies incorporate train event variables when developing a train delay predictive model, followed by location and external variables. However, very few studies consider train and infrastructure variables. The availability of data and the variables' relevance are key considerations when deciding which variables to be incorporated into a predictive model. However, one critical gap in previous studies for train event variables has been identified. Existing factor analysis studies consider the impact of delays in previous stations for the same train but ignore the delay impact of ahead trains passing the same station (reflecting interaction between trains) or train delays from forward stations in previous times (reflecting traffic conditions). These variables are expected in improving train delay predictions and understanding their impacts would be beneficial for better traffic controls. To gain a more in-depth understanding of the relationship between various factors and train delays, the paper examines the impact of a comprehensive set of factors, including train operations, railway networks, maintenance, weather calendar, and train interaction and traffic conditions.

3. Methodology

3.1. Problem formulation

Most existing studies on understanding the factors' impacts on train delays model the problem using a single, general equation as Eq. (1).

$$y_i = f(\mathbf{X}, \boldsymbol{\beta}) \quad (1)$$

where y_i is the arrival delay of the train at station i , $\mathbf{X}$ is a set of explanatory variables, and $\boldsymbol{\beta}$ is the set of parameters to be estimated.

However, Eq. (1) cannot capture the heterogeneous impacts of spatiotemporal variables on train delays as the train travels downstream along the train lines. For example, the consecutive upstream station delays may have different impacts on the current train delays depending on where the train is and how the operation is scheduled at these stations. In this paper, we model the real-time train arrival delay factors analysis model as a sequence of regression equations conditional on where the train is, mathematically:

$$y_i = f(\mathbf{X}_i, \boldsymbol{\beta}_i \mid \text{train is at station } i - 1), \forall i \in [1, I] \quad (2)$$

where the train arrival delays y_i at station i is a function of explanatory variable set $\mathbf{X}_i$ conditional on the train is at station $i - 1$. $\boldsymbol{\beta}_i$ are the parameters to be estimated corresponding to the delay at station i .

Compared to Eq. (1), Eq. (2) is more flexible in incorporating different sets of variables $\mathbf{X}_i$ in estimating delays at certain stations. In addition, Eq. (2) can capture the heterogeneous impact of variables by varying the coefficient $\boldsymbol{\beta}_i$. Eq. (2) is a set of equations with the size of I for a train line with $I + 1$ stations, which is expanded as:

$$\begin{aligned} y_1 &= \mathbf{X}_1 \boldsymbol{\beta}_1 + \varepsilon_1, \\ y_2 &= \mathbf{X}_2 \boldsymbol{\beta}_2 + \varepsilon_2, \\ &\vdots \\ y_I &= \mathbf{X}_I \boldsymbol{\beta}_I + \varepsilon_I \end{aligned} \quad (3)$$

where ε_i is the error terms.

To ensure the generality of results, many studies that analyze factors affecting train delays tend to combine data from multiple routes (Palmqvist et al., 2017; Gorman, 2009). However, this approach neglects the potential unexplained heterogeneity that exists across different routes by treating all observations independently. The proposed model estimates the arrival delays for the next station conditional on the current train location at the trip level. Thus, the model has an advantage over the commonly adopted approach in capturing the heterogeneity of trip-level factors that influence train delays. For example, the impact of the previous station delay on the current station delay for route 1 would be different from these for route 2.

The proposed model has the forecasting capability in the context of real-time operations and information provision, but not for the planning application. Under such an application context, the prediction will be made for the next station when a train arrives at the current station. As illustrated in Fig. 1, this approach involves predicting next trip arrival delays $d_{k,i+1}$ given train location i at current time t . The model incorporates observed information at the time of prediction t , allowing for the exploration of the heterogeneous impact of various explanatory variables on next-trip arrival delays given the most recent information. It should be noted that one challenge for this real-time prediction context is the observability level of information, which means that only factors that can be observed in real-time and historically, such as initial delays at terminal stations, current train delays, and historical train delays, are investigated for their impact on next trip arrival delays. In other words, the proposed model will not be viable if the train journey has not commenced, i.e., if there is no initial delay. However, by adopting the concept of a real-time prediction approach, the contribution of each variable to the train delay can be explored in real-time, providing a better understanding of trends in subsequent train arrival delays as they respond to changes in the operating environment. To the best of the author's knowledge, the exploration of the factors that influence train arrival delays from a real-time perspective has not been done before.

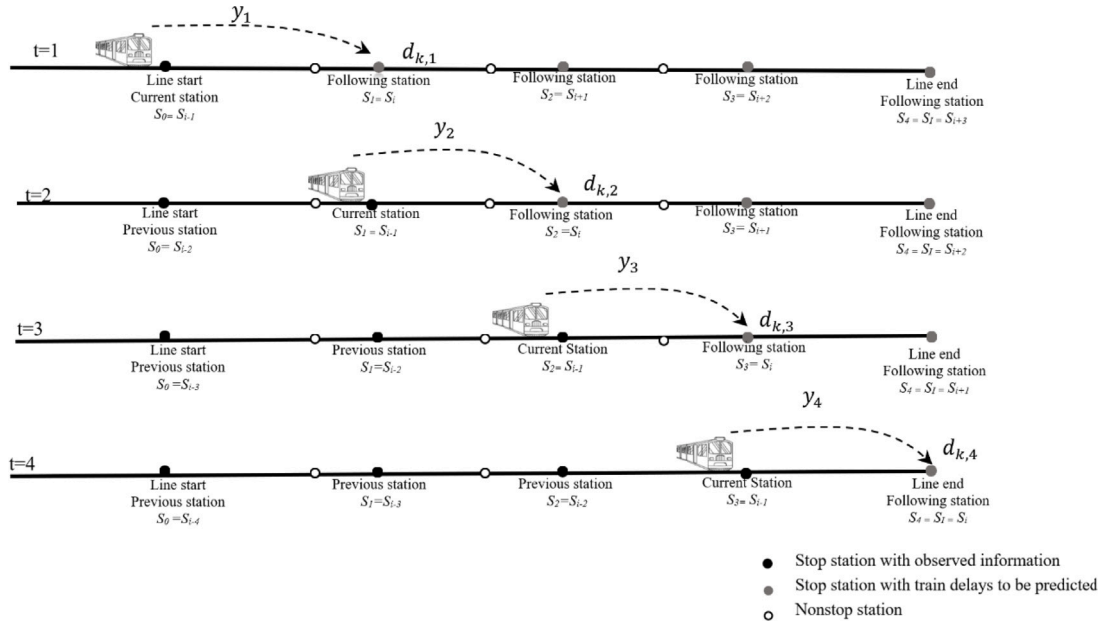


Fig. 1. Real-time train arrival delay factors analysis framework.

3.2. Seemingly unrelated regression equation model

To estimate parameters in Eq. (3), the traditional OLS method could be applied independently to each regression equation. Although each equation in Eq. (3) does not directly interact with one another, the contemporaneous disturbance term correlations are expected to exist across them caused by the commonly missed feature variables (e.g., driver behavior). This is because these equations are for estimating arrival delays of the same train traveling along the same line toward the same final destination. It is impossible in practice to collect or measure all possible variables that affect train arrival delays, thus the error terms in these equations may share the same patterns, leading to contemporaneous correlations among equations. In other words, although these equations are seemingly unrelated, they actually have unobserved properties in common and ought to be treated as a group for their parameter estimation.

Based on Eq. (3), the general model structure for the real-time train arrival delay factors analysis model is illustrated in Eq. (4).

$$y_i = \mathbf{X}_i \beta_i + \varepsilon_i, \quad i = 1, \dots, I \quad (4)$$

where y_i is a $N \times 1$ arrival delay of the train at station i with $n = 1, \dots, N$ is the individual observation, X_i is a $N \times k_i$ full column rank matrix of explanatory variables, β_i is the $k_i \times 1$ vector of regression coefficients, and ε_i is the $N \times 1$ error terms independent of X_i .

If the parameters of each equation in Eq. (4) are estimated separately by OLS, the error terms are implicitly assumed to be contemporaneously uncorrelated with the error structure as in Eq. (5), and any potential correlation between the equations is completely disregarded.

$$E[\varepsilon_i] = 0 \text{ and } E[\varepsilon_i \varepsilon_i'] = \sigma_i^2 I_N \quad i = 1, \dots, I \quad (5)$$

where $i \neq i'$

The SURE model is proposed in this study to overcome the drawbacks of OLS and solve the real-time train arrival delay factors analysis problem. SURE is a system of linear equations, with each equation in the system having its own parameter vector β_i , as shown in Eq. (4), making them unrelated, but the contemporaneous correlation across the error terms exists as in Eq. (6).

$$E[\varepsilon_i | X_1, X_2, \dots, X_I] = 0 \text{ and } E[\varepsilon_i \varepsilon_i' | X_1, X_2, \dots, X_I] = \sigma_{in} I_N \quad (6)$$

Under this assumption, the overall variance-covariance matrix of the stacked error terms is

$$E[\varepsilon \varepsilon'] = \begin{bmatrix} \sigma_1^2 I_N & \sigma_{12} I_N & \dots & \sigma_{1I} I_N \\ \sigma_{21} I_N & \sigma_2^2 I_N & \dots & \sigma_{2I} I_N \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{I1} I_N & \sigma_{I2} I_N & \dots & \sigma_I^2 I_N \end{bmatrix} = \Sigma \otimes I_N \quad (7)$$

where Σ is a positive symmetric matrix and I_N is a $N \times N$ identity matrix and $\otimes$ denotes Kronecker product.

The SURE model makes the assumption of the common disturbance terms for the model estimation, which fits for our problem model characteristics (trip-level regression conditional on the station). That is because, at the trip level, the possible common missing factors (disturbance terms) are basically the same, such as rolling stock, tracks, driver styles, etc. The utilization of SURE accounts for the presence of contemporaneous correlation across the trip-level disturbance terms, which provides an advantage over OLS (disregarding such correlation).

Several studies applied the SURE model to understand the impact of explanatory variables on the response variables in areas of public transport reliability and travel behavior analysis (Ma et al., 2015; Nasri and Zhang, 2019; Sheng and Sharp, 2019). One main advantage of utilizing SURE to estimate the train delay factors is that it allows each regression model to benefit from the information contained in other regression equations by accounting for correlated errors among different equations caused by unobserved characteristics. The theoretical discussion of the SURE model could refer to Washington et al. (2020), Wang et al. (2011), Fraser et al. (2005).

According to Tiong et al. (2023), train delay prediction models can be broadly categorized into long-term and short-term prediction models. The former is used to derive insights and aid in the understanding of the impact of endogenous and exogenous factors on railway system performance, while the latter focuses on the accuracy of prediction models in the real world. The model proposed in this study falls under the long-term prediction model category. As the primary objective of this study is to understand the factors influence for real-time train delays at the trip level (not building prediction model itself), a simpler and more interpretable SURE model is selected over a nonlinear model. The SURE model enables statistical inferences and assesses the uncertainty of parameter estimates to provide valuable insight into the relationships between explanatory variables and train delays. In contrast, nonlinear models such as machine learning models often generate complex and less interpretable outcomes, although they are known for their strong predictive capabilities.

3.3. Data and model variables

Since the study focuses on investigating the factors influencing real-time train delays, the selection of explanatory variables is based on previous literature on train delay prediction (see Section 2.2), combined with data accessibility considerations. The study ultimately uses three data sources, namely train operation, trackwork, and weather data to extract the variables that could potentially contribute to train delays.

The train operation data is provided by the Swedish Transport Administration, Trafikverket. The data, which is recorded by the signaling system, details the scheduled and actual arrival and departure times of different trains at each station along the designated train path in Sweden, with one-minute time precision. The track utilization plan, obtained from Trafikverket's internal database, specifies the scheduled trackwork identification numbers, as well as their times and locations, to keep track of temporary capacity restrictions for Swedish train traffic. The weather data is provided by the Swedish Meteorological and Hydrological Institute (SMHI). The dataset includes historical meteorological observations of average snow depth, precipitation, and temperature recorded once per day, as well as maximum wind speed recorded hourly across Sweden. Due to the similarity in the data structure, the three datasets are merged based on the location and date-time of the train trip.

Table 3 lists the response and explanatory variables and their definitions in the model. The response variable is the arrival delays of a train at a station. The selected and designed explanatory variables include train operations, network, calendar, weather, and maintenance variables.

4. Case study and data preparation

4.1. Selection of study areas

This study evaluates train arrival delays along two high-speed train (HSR) lines: Copenhagen Central Station (KPH)–Stockholm Central Station (CST) and CST–Sundsvall Central Station (SDC). The rationale for having two rather different routes to support the generality of our findings. It is worth noting that the case study includes two different train lines in both directions, which effectively represents four distinct train lines with different traveling characteristics (KPH to SDC, SDC to KPH, SDC to CST, and CST to SDC). These train lines are chosen because they are frequently used, thus providing sufficient data for the analysis. The inclusion of multiple train lines is to ensure that the selected cases are representative and diverse enough to provide meaningful insights into the research question. That is, to capture the heterogeneous impact of various variables, as stated in Table 3, on train delays. Also, the four train lines selected are representative of the wider population of train lines in Sweden, as they cover long distances and are subject to a variety of environmental and operational factors. There are no HSRs scheduled to run from KPH to SDC because the entire journey takes about one day, resulting in inefficient rolling stock utilization. Fig. 2 shows the layout of HSR lines studied (see the red line with yellow dots in Fig. 2). Data for HSRs traveling along the chosen train routes between December 2017 and December 2020 is selected for analysis.

4.2. Data preprocessing and model estimation

In order to maintain consistency and capture the impact of various factors on train delays throughout the line as the train travels downstream, five stations from each line are selected. These stations are Hudiksvall, Söderhamn, Gävle, Uppsala, and Arlanda for

Table 3
Description of variables.

Variables	Type	Unit	Description
Response Variables			
$d_{k,i}$	Continuous	min	The arrival delay of train k at station i , the difference between the actual arrival time $t_{k,i}^a$ and the scheduled arrival time $t_{k,i}^s$ of train k at station i .
Exploratory Variable			
(1) Train operation variables			
$dw_{k,i-1}^s$	Continuous	min	Scheduled dwell time at the consecutive upstream station, that is the difference between the scheduled departure time $t_{k,i-1}^d$ and the scheduled arrival time $t_{k,i-1}^s$ of train k at the consecutive upstream station $i-1$ of station i .
$dw_{k,i-1}^a$	Continuous	min	Actual dwell time at the consecutive upstream station, that is the difference between the actual departure time $t_{k,i-1}^d$ and the actual arrival time $t_{k,i-1}^a$ of train k at the consecutive upstream station $i-1$ of station i .
$r_{k,i}^s$	Continuous	min	Scheduled running time between station $i-1$ and i , that is the difference between scheduled arrival time $t_{k,i}^s$ at station i and the scheduled departure time $t_{k,i-1}^d$ at its consecutive upstream station $i-1$.
$r_{k,i-1}^a$	Continuous	min	Actual running time between station $i-2$ and $i-1$, that is the difference between the actual arrival $t_{k,i-1}^a$ at consecutive upstream station $i-1$ and the actual departure time $t_{k,i-2}^d$ at its preceding station $i-2$.
$d_{k,j}, \forall j \in [1, i-1]$	Continuous	min	A set of explanatory variables of train arrival delays at upstream stations before station i . It varies with sizes depending on which station arrival delay to predict.
$d_{g,i}, \forall g \in [k-3, k]$	Continuous	min	A set of explanatory variables indicating the arrival delays of three train before train k at station i .
$\bar{d}_{k,i}^{hist}$	Continuous	min	Historical mean for arrival delays of train k at station i during the same time-of-day interval $\bar{d}_{k,i}^{hour}$, weekday $\bar{d}_{k,i}^{week}$ and month $\bar{d}_{k,i}^{month}$
(2) Network variables			
Direction	Categorical	–	Northbound: Direction of train traveling from KPH-CST or CST-SDC ; Southbound: Direction of train traveling from CST-KPH or SDC-CST
(3) Calendar variables			
Time of day	Categorical	–	Morning peak: [6 am, 10 am]; Afternoon peak: [4 pm, 9 pm]; else Non-peak
Days of week	Categorical	–	Weekdays: [Monday, Friday]; Weekends: [Saturday, Sunday]
(4) Weather variables			
Temperature	Categorical	–	Normal temperature: Temperature > 0 °C; Cold: Temperature $[-5$ °C, 0 °C] ; Very cold: Temperature < -5 °C;
Precipitation	Categorical	–	Light rain: Precipitate [0 mm, 10 mm]; Rain: Precipitate > 10 mm; else No rain
Snow	Categorical	–	Have snow: Snow depth > 0 cm; else No snow
Wind	Continuous	–	Calm wind: Maximum wind speed [0 m/s, 13 m/s] ; Strong wind: Maximum wind speed > 13 m/s ; else No wind

(continued on next page)

Table 3 (continued).

Variables	Type	Unit	Description
(5) Maintenance variables			
Trackwork Period	Categorical	–	Daytime: Trackworks take place in the daytime; Nighttime : Trackwork take place in the night-time; Both: Trackworks take place both in the daytime and night-time; else No trackwork
Trackwork Restriction	Categorical	–	Have restriction: Speed restriction due to trackworks; else No restriction



Fig. 2. Swedish Southern Mainline.

the CST to SDC line and Hässleholm, Alvesta, Nässjö, Linköping, and Norrköping for the KPH to SDC line. The selection of these stations is based on the availability of data and the sufficiency of coverage for long-distance trains passing through the stations, as the number of long-distance trains passing through these stations is limited. Furthermore, runs other than HSRs are excluded from the analysis; however, the arrival delays of earlier trains $d_{g,i}$ are computed based on the complete set of trips. It should be noted that one of the delay-related variables, the historical mean for arrival delays at the current station, $\bar{d}_{k,i}^{month}$, $\bar{d}_{k,i}^{week}$, and $\bar{d}_{k,i}^{hour}$ are computed based on data from December 2016 to December 2020. These historical arrival delays, $\bar{d}_{k,i}^{month}$, $\bar{d}_{k,i}^{week}$, and $\bar{d}_{k,i}^{hour}$, are average arrival delays for the same month of the year, the same weekday, and the same hour of the day, respectively. The calculation of the remaining variables is self-explanatory, as shown in Table 3.

A variety of data cleaning and data transformation tasks were implemented to organize the inputs for the SURE model. The data has over 10,000 trips on the study route in four years. The archived data is processed by removing incomplete journey records and observations with extreme or abnormal values (exceeding 2 standard deviations from the mean value). After filtering, the used model data contains a total of 46,875 train records.

To address the non-linear relationship between the explanatory variables and train delays, the discretization approach is used based on their impacts on train delays by conducting a series of trial-and-error tests. The same method is used by Økland and Olsson (2021) who categorized the weather variables as they hypothesized that weather negatively influences train delays and punctuality when snowfall, precipitation, and low temperatures hit certain thresholds. After a series of experiments (maximizing the model R^2), the explanatory variables, including calendar, weather, and maintenance variables, are grouped into distinct categories, as shown in Table 3, resulting in a more precise capture of non-linear relationships. The categorical variables are then one-hot encoded, that is, dummy variables are created. To prevent perfect collinearity, the baseline feature, one dummy variable for each categorical variable, is excluded. The variables are then all subjected to a Pearson correlation analysis to check for multicollinearity. When two variables are highly correlated, the more significant one is used. The models are estimated via SURE using the SUR function under the systemfit package in R-studio.

4.3. Model results and analysis

Table 4 summarizes the estimation results of the SURE model for arrival delays. The relatively low RMSE within 4 min and the high R^2 indicate the high performance of the SURE model for further analysis. The R^2 values show that the model can explain between 83% and 89% of the variability of train arrival delays. Compared to a close study using OLS regression models by Palmqvist et al. (2017) (R^2 is 0.043), the proposed local models through conditional regressions perform significantly better than a global model. In addition, we ran our models using the random forest, and the model R^2 (0.91) is comparable to our SURE model results (0.88–0.9). We believe it is more reasonable to use the SURE model by tradeoff the interpretability and accuracy of the model. Table 4 also shows that the model performance improves as the train moves along the route toward the destination station since more real-time train operation information can be used as explanatory variables.

4.3.1. Impact of train operation variables

The scheduled dwell time at the preceding station $dw_{k,i-1}^s$ has a significant negative impact on current station arrival delays. This is expected because a longer scheduled dwell time on the station will absorb delays and give more flexibility for delay recovery by shortening the dwell time. The absolute negative coefficients in the beginning sections of lines tend to be higher. It suggests that larger scheduled dwell times could be allocated at the initial and middle stretches of the line to prevent the early initiation of train delays, which could propagate to stations further downstream and affect more stations.

The actual dwell time at the previous station $dw_{i-1,k}^a$ is statistically significant and positively correlated with current arrival delays. This finding is consistent with existing studies in Yuan and Hansen (2007), Harris et al. (2013). They found that station dwell time is the major source of train delays. The train with longer dwell times, which are primarily brought on by passengers' boarding and alighting (Watanabe et al., 2018), can induce departure delays and increase the risk of delayed train arrivals at the subsequent station. In general, we can observe that the dwell time impact is higher for delays when a train approaches the destination terminal stations. This could be because passengers who boarded at earlier stations would be more likely to alight at stations further downstream. The lower absolute negative coefficient of scheduled dwell time at the end of the line also suggests that the high volume of passenger alighting activity at the last stations could have been taken into consideration when constructing the timetable.

Current arrival delays and the scheduled running time for current section $r_{i,k}^s$ have a positive relationship in the first sections of lines, and then followed by a negative relationship. The train delays contain both positive and negative values, representing late and early train arrivals at stations, respectively. The findings imply that longer scheduled running times at the first station will result in earlier train arrivals, whereas increasing the scheduled running times at later stretches will decrease train arrival delays. It has significant policy implications for optimal capacity utilization. For example, it would be better to dedicate more scheduled running times to the later stretch of the train line in order to absorb accumulated delays and give more flexibility for delay recovery by adjusting the train speed. This is because delays experienced in the beginning are relatively minor, and the unused scheduled time will be wasted because trains cannot depart earlier, requiring early-arriving trains to dwell longer to prevent early departures.

The actual running time for the previous section $r_{i-1,k}^a$ has a negative impact, suggesting that the early arrival is prolonged with a short actual running time from the previous line section. The results suggest that a short actual running time following an overtaking operation leads to early train arrival at the station, causing the trains to queue in order to use train infrastructure such as station platforms or tracks due to the limitation of infrastructure capacity. This implies that it would be beneficial to control overtaking or speeding in train operations because the arrival time of trains could have a greater impact on congestion delays than their count (Gibson et al., 2002).

4.3.2. Impact of operation delay variables

As expected, current station delays increase with the delay at the consecutive upstream station $d_{k,i-1}$. This reflects that the delays from the consecutive upstream station $d_{k,i-1}$ potentially aggravate the current station delays due to its direct propagation to the current station. Interestingly, the second preceding station arrival delay $d_{k,i-2}$ has a negative but weaker relationship with the current station delays. In other words, the second preceding station's arrival delays may result in shorter arrival delays at the current stations. This may be due to dispatchers having more time to make necessary adjustments, like adjusting train speeds, to catch up with schedules when they are given earlier notice that the trains are running behind schedule from the second preceding station. Overall, the effect of $d_{k,i-1}$ is consistently significant and greater than that of the further upstream stations (i.e., $d_{k,i-2}$, $d_{k,i-3}$, $d_{k,i-4}$). The findings demonstrate that the consecutive upstream station delay of the same train has a considerable impact on the current station train delay, while the impact of delays from further upstream stations is marginal. That could be because the impacts of further upstream delays are absorbed by the consecutive upstream station delay. Thus, information on delays at the nearest station (i.e., the consecutive upstream stations) is more reflective of current traffic conditions and thus more crucial for real-time operations control and driver adjustments.

The impact of delays from ahead trains that pass the same station (i.e., $d_{k-1,i}$, $d_{k-2,i}$, $d_{k-3,i}$) consistently shows a statistically significant positive impact on the current station delays. It indicates that the interaction between trains consistently has a significant impact on the current station arrival delays. This is because these preceding trains share the same infrastructure as the current train, and any exceedance of the scheduled infrastructure occupation time by any of the trains will cause hindrances to the current train operation or even cause route conflicts. Based on this analysis, it is suggested that measures to control the punctuality of all trains, that is, ensuring all trains arrive on time at the stations, would be generally advantageous for operations.

Table 4
Summary of SURE model estimation results.

Parameter	Coefficient ^{p-value} for $d_{k,i}$			
	$d_{k,1}$	$d_{k,2}$	$d_{k,3}$	$d_{k,4}$
Intercept	-3.144***	—	—	1.257***
(1) Train operation variables				
$dw_{k,j-1}^s$	-0.933***	-1.140***	-0.816***	-0.601***
$dw_{k,j-1}^a$	1.145***	1.188***	1.238***	1.406***
$r_{k,i}^s$	0.072***	—	-0.049***	-0.054***
$r_{k,i-1}^a$	N.A.	-0.047***	-0.016*	-0.075***
$d_{k,j-1}$	1.042***	1.079***	1.059***	1.072***
$d_{k,j-2}$	N.A.	-0.013*	-0.020*	-0.017**
$d_{k,j-3}$	N.A.	N.A.	0.017**	-0.018**
$d_{k,j-4}$	N.A.	N.A.	N.A.	0.015**
$d_{k-1,i}$	0.033***	0.037***	0.010***	0.029***
$d_{k-2,i}$	0.012***	0.017***	0.006***	0.020***
$d_{k-3,i}$	0.016***	0.015***	0.002*	0.023***
$\tilde{d}_{k,j}^{month}$	—	-0.043***	-0.065***	-0.021*
$\tilde{d}_{k,j}^{week}$	0.144***	0.050***	0.088***	0.123***
$\tilde{d}_{k,i}^{hour}$	0.048***	0.022**	0.105***	0.057***
(2) Network variables				
Direction (baseline: Southbound)				
(3) Calendar variables				
Time of day (baseline: Non-peak)				
Morning peak	—	0.086*	0.277***	—
Afternoon peak	0.136***	—	-0.095*	—
Days of week(baseline: Weekends)				
Weekdays	0.169***	0.200***	0.320***	0.190***
(4) Weather variables				
Temperature(baseline: Normal temperature)				
Cold	-0.151***	—	—	-0.159***
Extremely cold	-0.143***	0.095*	0.176***	-0.318***
Precipitation(baseline: No rain)				
Light rain	0.068*	—	—	—
Rain	0.129*	—	0.216**	—
Snow(baseline: No snow)				
Have snow	—	—	-0.305*	—
Wind(baseline: Calm wind)				
Strong wind	—	—	—	—
(5) Maintenance variables				
Period of trackwork (baseline: No trackwork)				
Day	0.592***	—	—	—
Night	—	—	—	—
Both	1.042**	—	2.142**	1.081**
Speed restriction (baseline: No restriction)				
Have restriction	-0.610*	—	0.668*	0.962***
RMSE	3.111	3.534	3.850	3.575
R2	0.828	0.833	0.842	0.889
Adj R2	0.828	0.833	0.842	0.889

Note: Positive coefficient indicates the parameter contributes to an increased train delay.

negative coefficient indicates the parameter contributes to a reduced train delay.

$d_{k,1}$ = Train arrival delays at the second station of each of the four lines.

$d_{k,2}$ = Train arrival delays at the third station of each of the four lines.

$d_{k,3}$ = Train arrival delays at the fourth station of each of the four lines.

$d_{k,4}$ = Train arrival delays at the fifth station of each of the four lines.

The four lines include KPH to SDC, SDC to KPH, SDC to CST, and CST to SDC.

N.A. = not applicable, — = insignificant variable with $p > .05$.

p -value = t -statistics significance, where

* = $p < 0.05$,

** = $p < 0.01$,

*** = $p < 0.001$.

The current station train delays have a significant positive relationship with the near-term historical delays at the same station (i.e., $\bar{d}_{k,i}^{week}$ and $\bar{d}_{k,i}^{hour}$), demonstrating that the current station delays are significantly higher than the near-term historical delays. This is probably due to the increase in train traffic volume and heterogeneity to accommodate the increasing passenger demand after the Covid-19 restriction is lifted. The longer-term historical delay (i.e., $\bar{d}_{k,i}^{month}$) can be viewed as a more general delay indicator and is less prone to temporary fluctuations in delays. Since the monthly average delay has overestimated the current station delays given an increase in $\bar{d}_{k,i}^{month}$ resulted in a decrease in current station delays, indicating an improvement in train service punctuality overall. This might be the case because train traffic volume has not completely returned to pre-Covid-19 levels. Thus, the identification of effective traffic management and dispatching strategies is crucial to maintain high levels of rail operations efficiency even with the rebound in traffic volume and to guarantee passenger satisfaction.

4.3.3. Impact of network variables

Northbound trains show statistically significant and longer delays than southbound trains in both terminal stations. This means that the northbound train suffers from delays earlier than the southbound train and has more difficulty arriving on time at the final station. This is due to the better placement of margins on southbound trains as opposed to northbound trains. The distribution of margins along the Swedish Southern Mainline, which is also included in this case study, was analyzed by Andersson et al. (2011), who asserts that margins should be located in such a way that they can be used to quickly absorb various types of disturbances, specifically shortly after the stations where the most delays occurred. This can be seen for margins that are arranged for southbound trains, thus giving them a better chance to recover from the delays (Andersson et al., 2011).

4.3.4. Impact of calendar variables

Compared to non-peak hours, the morning peak hours cause additional train delays. The possible reasons could be due to a heavier passenger load during the morning peak hour, which increases dwell times. According to Kuipers et al. (2021), passengers' exchange activity can affect dwell times in a variety of ways, such as late arriving passengers extending dwell times by door holding, railway staff offering extensive passenger service by holding the train for passengers to board, and clustered boarding extending the time required to complete the alighting and boarding process. Interestingly, the afternoon peak shows an increase in train delays along the first stretch of the line, followed by an earlier arrival along the second stretch. This could be due to the variability of passenger volume at different sections of the line, particularly when passengers are more flexible in deciding when and where to travel after work. Less congestion inside stations due to lower passenger numbers on the later stretch of the line, combined with sufficient train capacity allocated to accommodate potential passenger demands at the afternoon peak, alleviates congestion delays at the end of the line. Thus, platform management such as effective information provision and the use of gates to reduce the counter flow of passengers might be needed to minimize the variance and dwell time duration at the station, especially during the morning peak hour. To alleviate peak-period congestion by balancing the utilization rate of in-service trains, Liang et al. (2022) proposed adjusting train fares to change passenger travel time and reasonably loading the passenger flow onto each train during peak periods.

Weekday operations are more likely to result in an increase in train delays than the weekend. This is due to fewer trains being scheduled per day on weekends compared to weekdays (Trafikanalys, 2021). The greater number of trains leads to smaller headways between trains, which in turn leads to a greater potential for congestion interference between trains (Gorman, 2009), that is, the earlier trains have the potential to cause hindrance for later trains and raise the risk of delays spreading to other trains. This is because short headways cause the train to be unable to catch up with train delays despite accelerating to reduce running time. Thus, conscious timetable planning to minimize interaction between trains through appropriate capacity utilization and train spacing variability are important factors to account for, especially when the volume of train traffic is increasing to satisfy the increasing demand (i.e., during weekdays).

4.3.5. Impact of weather variables

The impact of temperatures on train delays varies along the line. This might be due to the fact that Sweden is a sizable country with huge discrepancies in temperature. At both terminal stations, train delays are reduced due to the cold and extremely cold temperatures. One possible explanation is that there are fewer passengers at the access points due to reduced train demand in the cold weather. According to Cools et al. (2010b), passengers are more likely to cancel or postpone their non-mandatory trips (i.e., leisure trips) during extreme temperatures. On the other hand, middle stations experience train delays due to the extremely cold temperature. This could possibly be due to the vulnerability of train operations in this section to disturbances during extremely cold weather. Train derailments, mechanical switch jams caused by ice formation, brittle tracks, railway track cracking, frozen overhead lines, and malfunctioning signaling equipment are examples of train disturbances in cold temperature conditions that can result in train delays due to failures of train facilities or reduce train speed for safety reasons. Hence, a comprehensive analysis is necessary to develop an effective strategy to address temperature variation across the line while taking local climate conditions into account.

Overall, precipitation is found to increase train delays. A potential reason is that driving in wet conditions is riskier, and this leads to an increased headway distance that needs to be preserved between trains, implying a reduced capacity of the railway network. In other words, the reduction in train speed due to precipitation increases the travel time required to complete the trips, resulting in more train congestion delays. Another reason could be that precipitation increases demand for public transportation (Aaheim and Hauge, 2005). This reasoning is strengthened by Creemers et al. (2015) who argued an increased use of public transport with trip distance due to limited alternative availability. As a result of the increased train demand, longer waiting times are required for passengers to board and alight the trains. A related consideration in this adverse weather condition is the potential increase in passengers missing connections due to train delays on parts of their journeys. Thus, to ensure the success of integrated transportation

services, it is critical to ensure the robustness of the timetable under adverse weather conditions by including dwell time slacks at transfer points.

The impact of snow on train delays are generally insignificant. This probably reflects the robustness of trains and train infrastructure to operate even in snowy conditions, potentially making railway transport an appealing mode of transportation during the snow. Interestingly, the only statistically significant snow coefficient shows that snow shortens train delays. The possible reasons are the changes in travel behavior caused by snow. Cools et al. (2010a), Cools and Creemers (2013) indicated that snowy weather is most likely causes trip postponements and trip cancellations for nonobligatory activities such as shopping and visiting activities; for obligatory activities such as school and work, passengers may try to avoid traffic congestion by altering their departure times or taking a different route. Although there will still be regional variations, it is worth noting that due to climate change, more precipitation will fall as rain than snow, even in the winter, resulting in lower levels of snow depth and shorter periods of snow cover in some areas of Sweden (Sweden Commission On Climate and Vulnerability, 2007). Thus, the adaptation measures to prioritize are those that address the risk of damage from increased precipitation on account of climate change.

Strong wind has a statistically insignificant impact on train delays when compared to calm wind. This seems to be unexpected, however, it can possibly be due to the magnitude of wind speed in the study area not being strong enough to have an impact on railway operating systems or influence the mode choice decision of passengers (i.e., shifting from other transportation modes to railways or vice versa). In general, the coefficients for weather variables have plausible signs, and the results are encouraging since the influences of most weather variables on train delays are mostly statistically insignificant. This suggests that Sweden's railway transportation system is reliable and less susceptible to weather factors. The findings enable policymakers to evaluate the appropriateness of traffic management strategies implemented during adverse weather conditions in order to maintain railway transportation as an attractive and reliable mode of transport under a variety of weather conditions.

4.3.6. Impact of maintenance variables

When compared to the no trackwork condition, the scheduled trackwork for both day and night consistently increases train delays, regardless of the train locations. This is as expected because the trackwork frequently causes speed reductions (van der Kooij et al., 2017) as well as capacity restrictions for train traffic due to track closures or single-track operations (Peterson et al., 2019). Moreover, train timetable rescheduling due to train path adaptation for trackwork is not always conducted (Ivina et al., 2021). The introduction of speed restrictions and capacity limitations in combination with unadjusted train timetables inevitably leads to unpunctual train services. The results also reveal that the trackwork scheduled to take place during both the day and the night has a greater tendency to cause an increase in train delays, followed by trackwork that is scheduled solely during the day. This implies that the greater the interference of scheduled trackwork with train movements, the greater the train delays. It is also worth noting that there are more trains operating during the day. Therefore, it is advised to execute maintenance work in periods when there is less traffic (i.e., at night) to minimize operational restrictions for regular train traffic caused by maintenance work.

Compared to stations with no temporary speed restriction due to track work, those with temporary speed restrictions due to trackwork show a positive correlation with train delays as the trains travel across the train line except for the station at beginning of the line. This could be because the initial station is less likely to be burdened with heavily accumulated delays from the previous station, giving it enough buffer time to deal with the temporary speed restriction. For the remaining downstream stations, the unadjusted timetable based on current speed limits depleted the initially scheduled buffers on the network, increasing the required travel time and thus causing an increase in train delays. As a result, the timetable rescheduling due to trackwork activities is required to redistribute the runtime margin and buffer time across the affected lines in order to increase timetable stability and prevent train delays from propagating to downstream stations.

5. Conclusion

To understand the heterogeneous impact of various factors on train delays, the paper models the train delay prediction problem as a set of next-station arrival delay prediction models conditional on the train location. We apply a seemingly unrelated regression equation (SURE) model to estimate the model parameters. The SURE model assured the statistically more efficient estimation of regression coefficients by taking into consideration the contemporaneous model residual correlations caused by the shared unobserved feature variables across regression equations, such as driver characteristics and train mechanical functionalities.

A case study was conducted for the high-speed rail lines in Sweden. The results confirmed the necessity of developing a set of station-specific delay prediction models as opposed to a single generic model. The heterogeneous impact of a comprehensive set of factors (i.e., train operations, network, weather, maintenance, and calendar variables) is observed for station delays. The empirical findings and implications for practice are summarized:

1. Train operation variables, such as dwell time, running time, and delays from previous trains and upstream stations, have significant impacts on train arrival delays at downstream stations. Considering the effects of these factors differ depending on the location of the station, different train operating management strategies should be targeted at different stations.
2. Calendar variables such as peak hours and weekdays (high volumes of passenger and rail traffic) have a positive impact on the current station delay. To reduce congestion during these periods, it would be beneficial to take into account strategies such as strategic timetable planning, effective platform management, and appropriate train fare adjustments.
3. Maintenance variables, such as the trackwork period and temporary speed restriction, have a significant positive impact on the train arrival delay. To reduce delays in train operations caused by maintenance activities, it would be useful to consider measures such as scheduling maintenance work during low-traffic periods or rescheduling operation timetables.

One generalized conclusion from the case study is that the factors' impact directions (positive/negative) are consistent while their impact levels vary depending on station locations. Based on these, generalized management recommendations could be made to reduce train arrival delays, for example, enhancing the terminal dispatch punctuality. Importantly, our model is generic and flexible to incorporate different sets of variables in estimating delays at certain stations and capture heterogeneous impacts of variables. The results acquired from the developed model can be used to drive policy decisions quantitatively and help policymakers make decisions to improve the punctuality of trains. As common in empirical studies, the interpretation of results is constrained to the case studies for the studied HSR line in Sweden. Future work will explore the practical application of variables that have been identified as significant in the real-time train arrival delay factors analysis. From a methodology perspective, more advanced learning-based models could be studied in the future to develop accurate real-time prediction models and to capture the complex interactions and non-linearity of complex relationships among explanatory and response variables.

CRedit authorship contribution statement

Kah Yong Tiong: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Zhenliang Ma:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing, Supervision. **Carl-William Palmqvist:** Resources, Data curation, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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